

BENARJEE SUDEEP SAMPATH PYLA

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RELEVANT EXPERIENCE

LexTrack AI (Capstone)

Berkeley, CA

Aug 2025 – Dec 2025

AI/ML Developer

- Built an end to end analytics pipeline using live LexTrack AI ecosystem data, handling data cleaning, feature engineering, model building, and workflow automation for business decision support.
- Developed predictive models and generative AI workflows to forecast key outcomes, surface trends, and turn complex data into clear recommendations for internal teams.
- Added data checks and explainable insights to make outputs more reliable, clear, and useful for stakeholders.

Astragen

Vijayawada, India

Dec 2023 – Apr 2024

Junior Data Engineer

- Designed and maintained the database architecture using MySQL to support a multimedia application used in schools, ensuring optimal performance and scalability for remote content access.
- Built usage/ops dashboards and standardized metric definitions to reduce reporting inconsistencies and improve stakeholder alignment.
- Developed pipelines for engagement tracking; partnered with engineers to validate data capture, syncing behavior, and correctness across systems

EVE HealthCare

Remote

Jun 2023 – Sept 2023

Data Analyst Intern

- Analyzed operational scheduling and capacity data to identify exception patterns and recurring bottlenecks, translating findings into process changes that improved planning decisions.
- Built KPI reporting in Power BI with standardized definitions and drilldowns, enabling leadership to monitor performance trends and investigate outliers faster.
- Quantified inefficiencies and translated results into clear recommendations, supporting project prioritization with numbers.

TECHNICAL SKILLS

Programming languages: Python, R, SQL

Data Science: Pandas, NumPy, scikit-learn, PySpark, TensorFlow Keras, time series, A/B testing

LLM and NLP: LangChain, FAISS, Hugging Face Transformers, OpenAI API, hybrid retrieval, RAG

Data Engineering: ETL pipelines, data modeling, data validation and reconciliation, schema design, REST APIs, JSON

Cloud and DevOps: Azure ADLS ADF Databricks Synapse, GCP Artifact Registry, Docker, Kubernetes, Helm, GitHub Actions

Databases and Messaging: MySQL, PostgreSQL, Redis, RabbitMQ

BI and Visualization: Power BI, Tableau, Streamlit, Matplotlib, Excel Google Sheets, Git

PROJECTS

Agentic AI Troubleshooting System (Ricoh Hackathon – 2nd Place) | Python, LangChain, OpenAI API, FAISS, Hybrid Retrieval

- Built an agentic AI troubleshooting system that decomposes complex technical queries, performs iterative multi-document retrieval, and synthesizes step-by-step solutions grounded in Ricoh technical manuals.
- Implemented a hybrid retrieval pipeline (semantic + keyword) with intelligent chunking and metadata tracking. Improved evidence accuracy and consistency across distributed documentation.
- Designed a citation-first generation workflow that verifies claims, maintains agent state, and produces fully traceable answers with document-level references, meeting strict no-hallucination constraints.

Cloud-Native Appointment System | React, FastAPI, Redis, RabbitMQ, Airflow, Docker, Kubernetes, GitHub Actions, GCP

- Built a microservices-based appointment platform for patient booking and doctor schedule management, improving scalability and clean service separation.
- Implemented JWT-based authentication and role-aware APIs, strengthening security and reducing unauthorized access risk.
- Deployed on GKE with Docker and Helm and automated CI/CD to speed up releases and keep deployments consistent.

Sentiment Analysis on Agnipath Scheme | Python, BiLSTM-CNN, XGBoost

- Developed a hybrid model using BiLSTM-CNN and XGBoost to analyze public sentiment on the Agnipath Recruitment Scheme via Twitter data.
- Achieved 84% accuracy in sentiment classification, improving model performance over traditional deep learning models
- Presented research findings at the 3rd International Conference on Advances in Data-driven Computing and Intelligent Systems (ADCIS 2024).

EDUCATION

University of Colorado Boulder, Colorado, USA

Aug 2024 – May 2026

Master of Science in Data Science | GPA: 3.91/4.00